

# Lab 3: Linux commands

In this lab, you will use Linux commands to learn how and when to use them. In this lab, you **MUST** create a directory “lab3/” and a text file “answers.txt” containing **ALL** the answers to the questions of this lab. You **do not need to present** this lab.

## 1) Users and permissions

We will create three users and assign them to groups having special permissions to allow them to share information. Because you are creating users in your system, you must use “sudo” to be able to do the work.

- Create three new users using **adduser**: teacher1 (fullname teacher1 with the password: 123456), student1 (fullname student1 with the password: 654321), student2 (fullname student2 with the password: 765432). Check that their home directory has been created.
  - a) In the text file “answers.txt”, put all the commands you used to get to the result.
- Create two groups: teachers and students. Add teacher1 to the group teachers; and student1 and student2 to the group students.
  - b) In the text file “answers.txt”, put all the commands you used to get to the result.
- In each student directory, create a directory “work/” and add the permissions to allow each owner to read, write and execute, the members of the teachers group to read and execute in this directory, and no permissions for the others (it means a teacher can access all the students “work/” directories but a students can only access their own “work/” directory and not the “work/” directory of other students). You must change the permissions and owners, groups of the /home/ directory and subdirectories to make this scenario happen. Check if everything is working properly.
  - c) In the text file “answers.txt”, put all the commands you used to get to the result.

## 2) Processes, Memory, Disks

We will now look at the system information regarding processes, memory, and disks. In a terminal, run the commands “firefox” and “xeyes” in the background of the terminal.

- What are the Linux commands to:
  - a) See only the processes running in this terminal?
  - b) See only the jobs (tasks) running in this terminal?
  - c) See the tree of processes (pstree) running in this terminal?
  - d) See all the processes with detailed information (aux) on your Linux system?
  - e) See the free memory on your Linux system, in bytes, MB, GB?
  - f) See the amount of disk space free on your system, in KB, in a human-friendly way?
  - g) See the amount of disk space used in total by your “/home/” directory, on your system, in KB, in a human-friendly way?

- h) Send the “SIGKILL” signal to xeyes?
- i) Send the “SIGKILL” signal to all the “firefox” processes?

### 3) System events, redirections, pipes

We will now look at some system information, use redirections and pipes to search text files.

- What are the Linux commands to:
  - a) List all the system messages (dmesg) from the boot of the system?
  - b) List the beginning lines of the system messages?
  - c) List the ending lines of the system messages?
  - d) Save the list of all the files (ls -la), including subdirectories, in the “/usr/lib/” directory into a “mylibs.txt” file in your home directory?
  - e) Count the number of lines of the file “mylibs.txt”? How many are there?
  - f) Add the list of all the files (ls -la), including subdirectories, in the “/usr/local/lib/” directory at the end of the “mylibs.txt” file in your home directory?
  - g) Count the number of lines of the file “mylibs.txt”? How many are there?
  - h) Find the library version/number after “libm.so” (the shared math library) in the file “mylibs.txt”? What is this number?
  - i) Find the file “xf86-video-intel-backlight-helper” in “mylibs.txt”, and display 5 lines before the result in the terminal? In which directory is this “xf86-video-intel-backlight-helper” file?
  - j) Count the number of shared libraries (named libsomething.so and libsomething.so.something) in mylibs.txt? How many are there?

### 4) Other commands...

We will now look at some other interesting (or not) commands...

- What are the Linux commands to:
  - a) List the environment variables of your Linux system?
  - b) List the PATH for your Linux commands?
  - c) Know the location of a command you enter in the terminal? For instance, how to know which “firefox” executable will be launched when you call “firefox&” from the terminal.
  - d) List information about your Linux version?
  - e) List advanced information about your CPUs?
  - f) List advanced information about your memory?

**This is the end. In this lab, you learned to use many linux commands to do many tasks.  
Congratulations, grattis!**